

Derivative of $y = e^{f(x)}$

For a differentiable function $f(x)$, the derivative of $y = f(x)$ is $y'(x) = f'(x)e^{f(x)}$.

Examples

1. $y = e^{x^2}$

2. $y = e^{3x^2}$

3. $y = e^{3x+1}$

4. $y = e^{5/x}$

5. $y = e^{\sqrt{x}}$

6. $y = e^{x \sin x}$



The Integral of $y = e^{f(x)}$

If $f(x)$ is a differentiable function of x such that $f'(x)$ is not a function of x then $\int e^{f(x)} dx = \frac{1}{f'(x)} e^{f(x)} + c$.

Examples

Evaluate the following.

1. $\int e^{2x+1} dx$

2. $\int e^{x/3} dx$

3. $\int x^2 e^{-x^3} dx$

4. $\int 3e^{x/2} dx$

5. $\int \frac{x}{2} e^{x^2} dx$

6. $\int e^{-x} dx$



Examples

1. Express each of the following as the sum or difference of logarithm.

(a) $\ln \cot x$

(b) $\ln \tan^2 x$

(c) $\ln (x^2 - 4)$

(d) $\ln \sqrt{\frac{x+1}{x-1}}$

(e) $\ln (3 \sin^x)$

2. Express the following as a single logarithm

(a) $\ln x + \ln y - 1$

(b) $\ln \left(\frac{e^3}{x}\right) + \ln \frac{x}{e}$



The Derivative of $\ln f(x)$

Given $y = \ln(f(x))$, $y' = \frac{f'(x)}{f(x)}$.

Example

Find the derivative of the following functions

1. $y = \ln(x^2 + 2)$

2. $y = \ln\left(\frac{x}{\sqrt{x^2+1}}\right)$

3. $y = \ln\left(\frac{x}{(x-1)^2}\right)$

4. $y = \ln(2 - x^3)^2$



Examples

1. Find $\frac{d}{dx}(5^x)$ and deduce $\int 5^x dx$
2. Evaluate $\frac{d}{dx}(x2^{x^2})$ and deduce $\int 2^{x^2} dx$
3. Evaluate the following integrals
 - (a) $\int 3e^{2x} dx$
 - (b) $\int x^2 e^{x^3} dx$
 - (c) $\int 2^{\tan x} \sec^2 x dx$



Example

Evaluate the following integrals.

$$1. \int \frac{1}{2x}, dx$$

$$2. \int \frac{1}{2x-1}, dx$$

$$3. \int \frac{x}{x^2+1}, dx$$

$$4. \int \frac{1}{2x+3}, dx$$

$$5. \int \frac{1}{1-x}, dx$$

$$6. \int \frac{x^2}{x^3-2}, dx$$

$$7. \int \frac{x}{x-1}, dx$$



Example

Evaluate the following

1. $\int_{-4}^{-3} \frac{1}{x} dx$

2. $\int_{-1}^{-1/2} \frac{1}{x} dx$

Example

Can any meaning be assigned to $\int_{-2}^2 \frac{1}{x} dx$



Examples

Find the following integrals

1. $\int x \cos x \, dx$

2. $\int x \sin x \, dx$

3. $\int x \cos 2x \, dx$

4. $\int x e^x \, dx$

5. $\int x \ln x \, dx$

6. $\int x^2 \sin x \, dx$

7. $\int x^2 e^x \, dx$

8. $\int x^2 \cos x \, dx$

9. $\int x \ln x \, dx$



Example

Find the following integrals

1. $\int x \ln x \, dx$

2. $\int e^x \ln x \, dx$

3. $\int x^2 \ln x \, dx$



Example

Find the derivatives of the following trigonometric functions.

1. $y = \cos 4x$

2. $y = \sin 4x^2 + 1$

3. $y = \tan x^2$

4. $y = \cos x + \sin x$

5. $y = 3 \cos 5x^2 + 2$



Example

Differentiate the following with respect to x .

1. $y = \sin^3 x$

2. $y = \cos^4 x$

3. $y = \tan^2 x$

4. $y = 2 \cos^2 (x^2 + 1)$

5. $y = \sin^2 (4x^3)$

6. $y = 3 \tan^4 x$

7. $y = x^4 \cos x$

8. $y = \tan^9 4x^2$



Examples

1. If $y = \tan^{-1}(x^2 + 1)$. Show that $\frac{dy}{dx} = \frac{2x}{x^4 + 2x^2 + 2}$

2. Differentiate with respect to x the following.

(a) $\cos^{-1} x$

(e) $\operatorname{cosec}^{-1} x$

(b) $\sin^{-1}(2x + 1)$

(f) $\sin^{-1} x^2$

(c) $\cot^{-1} x$

(g) $\cos^{-1}(2x - 1)$

(d) $\sec^{-1} x$

3. Find the following

(a) $\frac{d}{dx}(\sin^{-1} x + \cos^{-1} x)$

(b) $\frac{d}{dx}(\tan^{-1} x + \cot^{-1} x)$



Examples

Find the following integrals

1. $\int 3 \sec^2 x \, dx$

2. $\int \sin 4x \, dx$

3. $\int 3 \cos (4x - \pi/2) \, dx$

4. $\int (\sec^2 2x - \operatorname{cosec}^2 4x) \, dx$

5. $\int (\cos 3x + 3 \sin x) \, dx$

6. $\int 2 \sec^2 4x \, dx$



Examples

Find the integral of the following

1. $\int \sin^2 x \, dx$

2. $\int \cos^2 2x \, dx$

3. $\int \sin^4 x \, dx$

4. $\int \cos^4 3x \, dx$

5. $\int \sin^2 6x \, dx$



Examples

Integrate the following

1. $\int \sin^3 x \, dx$

2. $\int \sin^5 x \cdot 2x \, dx$

3. $\int \cos^3 4x \, dx$

4. $\int 3 \cos^3 (2x + 1) \, dx$

5. $\int \sin^3 6x \, dx$



Examples

Find the following integrals

1. $\int \sin 3x \cos^2 3x \, dx$

2. $\int \sec^2 x \tan^2 x \, dx$

3. $\int \sec^5 x \tan x \, dx$

4. $\int \cos x \sin^4 x \, dx$



Examples

Find the following integrals

$$1. \int \frac{3x}{\sqrt{2x^2 + 1}} dx$$

$$2. \int 5x\sqrt{2x^2 + 1} dx$$

$$3. \int \frac{x - 1}{(n + 1)^3} dx$$

$$4. \int x^2\sqrt{x^3 + 1} dx$$

$$5. \int \sin^4 x \cos 4 dx$$

$$6. \int x(3x^2 + 2)^4 dx$$

$$7. \int x(2x + 3)^{1/2} dx$$



Examples

Find the following integrals

$$1. \int \frac{1}{\sqrt{1-x^2}} dx$$

$$2. \int \frac{1}{1+x^2} dx$$

$$3. \int \frac{1}{\sqrt{9-4x^2}} dx$$

$$4. \int \frac{1}{\sqrt{4-x^2}} dx$$

$$5. \int \frac{1}{\sqrt{1-3x^2}} dx$$

$$6. \int \frac{1}{\sqrt{16-9x^2}} dx$$

$$7. \int \frac{1}{4+x^2} dx$$

$$8. \int \frac{1}{1+16x^2} dx$$

$$9. \int \frac{1}{3+4x^2} dx$$

$$10. \int \frac{2}{1+x^2} dx$$

$$11. \int \frac{3}{\sqrt{1-x^2}} dx$$

$$12. \int \frac{1}{\sqrt{25-x^2}} dx$$

$$13. \int \frac{3}{4+x^2} dx$$

$$14. \int \frac{-4}{16-4x^2} dx$$



Example

1. Prove the following identities

(a) $\cosh x - \sinh^2 x = 1$

(b) $1 - \tanh^2 x = \operatorname{sech}^2 x$

(c) $1 - \coth^2 x = \operatorname{cosech}^2 x$

(d) $\cosh x + \sinh x = e^x$

(e) $\cosh x - \sinh x = e^{-x}$

2. Use the definition $\sinh x = \frac{1}{2}(e^x - e^{-x})$ and $\cosh x = \frac{1}{2}(e^x + e^{-x})$ to show that

(a) $\cosh(-x) = \cosh x$

(b) $\sinh(-x) = -\sinh x$

(c) $\cosh x > \sinh x$



Examples

1. Show that $\int \frac{1}{\sqrt{x^2+a^2}} dx = \sinh^{-1} \frac{x}{a} + c$

2. Prove that $\int \frac{1}{\sqrt{x^2-a^2}} dx = \cosh^{-1} \frac{x}{a} + c$

3. Consider the integral $\int \frac{1}{a^2-x^2} dx$.

(a) Use partial fractions to show that

$$\int \frac{1}{a^2-x^2} dx = \frac{1}{2a} \ln \left(\frac{a+x}{a-x} \right) + c$$

(b) Use the substitution $x = a \tanh u$ to show that

$$\int \frac{1}{a^2-x^2} dx = \frac{1}{a} \tanh^{-1} \frac{x}{a} + c.$$



Examples

1. Find

$$(a) \int \frac{1}{\sqrt{x^2+2x+10}} dx$$

$$(b) \int \frac{1}{\sqrt{9+x^2}} dx$$

$$(c) \int \frac{1}{\sqrt{4+x^2}} dx$$

$$(d) \int \frac{1}{x^2+x+1} dx$$

$$(e) \int \frac{1}{\sqrt{x^2-6x+10}} dx$$

$$(f) \int \frac{1}{\sqrt{x^2+4}} dx$$

2. Use the substitution $e^x = \tan u$ to evaluate the following

$$(a) \int \operatorname{sech} x dx$$

$$(b) \int \cosh x dx$$

$$(a) \int \tanh x dx$$



Example

Solve the following equations

1. $2xy \frac{dy}{dx} + y^2 = e^2 x$

2. $x^2 \frac{dy}{dx} + 2xy = 1$

3. $x^2 \cos y \frac{dy}{dx} + 2x \sin y = \frac{1}{x}$

4. $e^y + xe^y \frac{dy}{dx} = 2$



Examples

1. Solve the differential equation $xy\frac{dy}{dx} + y^2 = 3x$
2. Find an integrating factor required to make the following differential equations into an equation and solve it
 - (a) $x\frac{dy}{dx} + 2y = e^{x^2}$
 - (b) $xe^y\frac{dy}{dx} + 2e^y = x$
 - (c) $2x^2y\frac{dy}{dx} + x^2y = 1$



Examples

1. Solve the differential equation $\frac{dy}{dx} + 3y = e^{2x}$; $y(0) = \frac{6}{5}$
2. Solve $\frac{dy}{dx} + y \cot x = \cos x$
3. Find the general solution of $\frac{dy}{dx} + 2xy = x$. Determine the particular solution with the initial condition $y(0) = -\frac{1}{2}$
4. Solve $x \frac{dy}{dx} + 2y = 2$



Case 1: Identical Real Roots

Example

1. Find the general solution of the differential equation
 $4y'' - 12y' + 9y = 0$.
2. Solve the following differential equation
 - (a) $\frac{d^2y}{dx^2} + 6\frac{dy}{dx} + 9y = 0$.
 - (b) $100y'' - 60y' + 9y = 0$.
 - (c) $y'' - 2y' + y = 0$; $y(0) = 0$ and $y(1) = e$



Case 2: 2 Distinct Real Roots

Example

Find the general solution of the following differential equation

1. $4y'' + 2y' - 15y = 0.$

2. $y'' = y.$

3. $y'' - 12y' + 20y = 0.$

4. $15y'' - 8y' + y = 0.$

5. $2y'' - 5y' - 3y = 0.$

6. $y'' - 25y = 0.$

7. $6y'' - y' - y = 0.$

8. $y'' - 6y' + 5y = 0; \quad y(0) = 1 \text{ and } y'(0) = 9$



Case 3: Complex Roots

Example

Solve the following differential equation

1. $y'' + 6y' + 10y = 0.$

2. $y'' - 2y' + 50y = 0.$

3. $y'' + 6y' + 34y = 0.$

4. $36y'' + y = 0.$

5. $y'' - 4y = 0.$



Case 1: $f(x)$ is a polynomial

Example

Solve the following differential equation

1. $y'' - 4y' + 3y = 9x + 6.$

2. $y'' + 4y' + 5y = 10x^2 + x.$

3. $y'' - 7y' + 12y = 4.$

4. $y'' - 4y' - 5y = 17 + 15x.$

5. $y'' - 6y' + 10y = 10x^2 + 18x - 6.$

6. $y'' + y' = 10.$



Case 2: $f(x)$ is an exponential function

Example

Solve the following differential equation

1. $y'' + 4y' + 5y = 10e^{-2x}$.

2. $y'' + y = 10e^{2x}$.

3. $y'' + 9y = 20e^x$.

4. $16y'' - 8y' + y = 27e^x$.

5. $y'' + 2y' + y = 16e^{-x}$.



Case 3: $f(x)$ is an trigonometric function

Example

Solve the following differential equation

1. $y'' - 4y' + 3y = 10 \sin 2x + 15 \cos 2x.$

2. $y'' + 6y' + 9y = 8 \sin 2x + 6 \cos 2x.$

3. $y'' + 3y' = 12 + \sin x.$



Example

Find the length of the given curve between the specified points or values of the parameter.

1. $y = x^3$; $A(0, 0)$ and $B(2, 8)$
2. $y = \sqrt{x}$; $A(0, 0)$ and $B(4, 2)$
3. $r = 5 \cos \theta$; $\theta_1 = -\frac{\pi}{2}$ and $\theta_2 = -\frac{\pi}{2}$
4. $x = e^t \cos t, y = e^t \sin t$; $t_1 = 0$ and $t_2 = 2\pi$
5. $x = 5 \cos t, y = 5 \sin t$; $t_1 = 0$ and $t_2 = 2\pi$
6. $x = at^2, y = 2at$; $t_1 = 0$ and $t_2 = 5$
7. $x = at^2, y = at^3$; $t_1 = 0$ and $t_2 = 4$

